

Einführung in die Kern-, Teilchen und Astrophysik

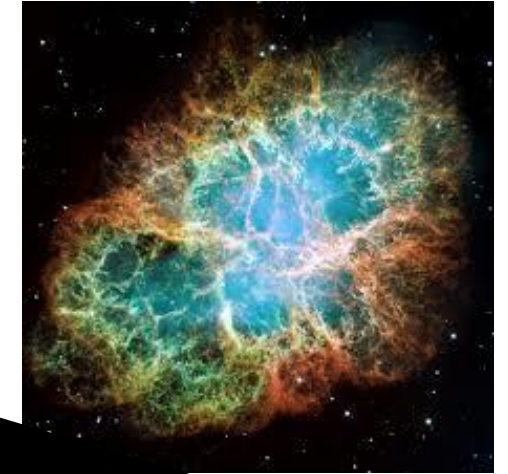
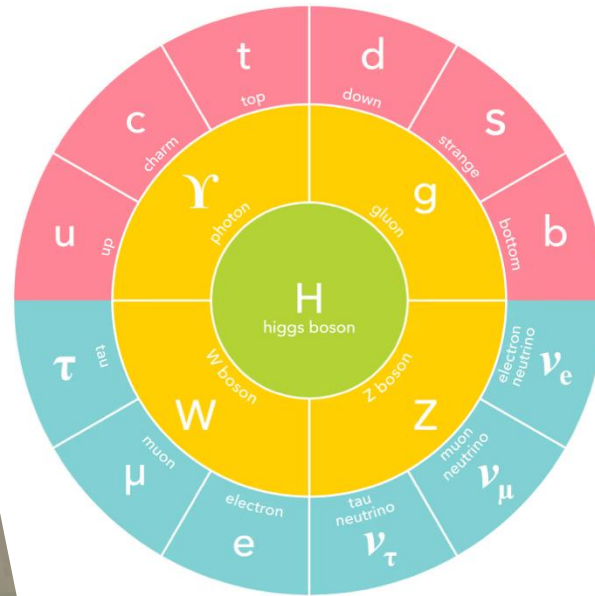
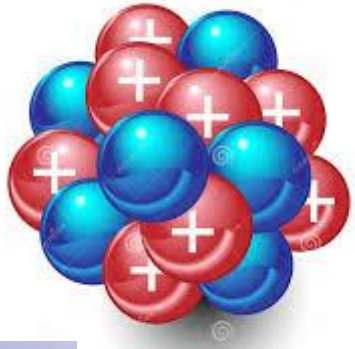
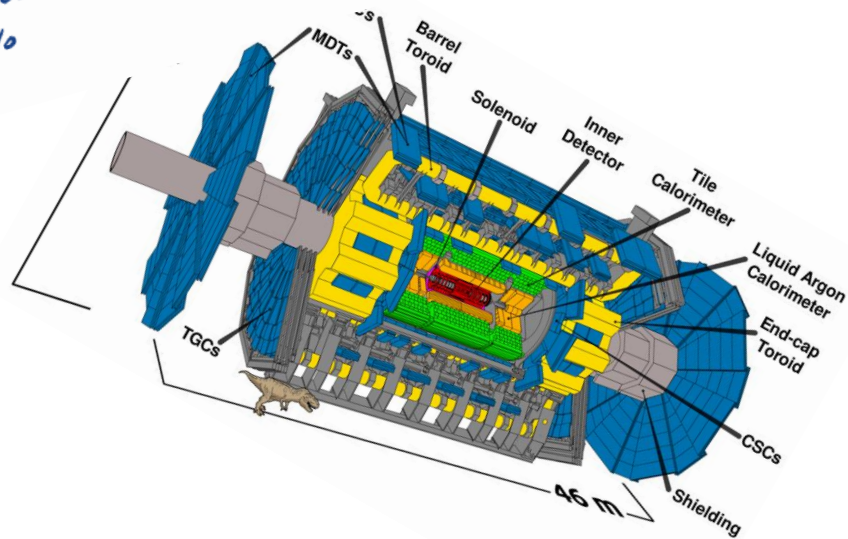
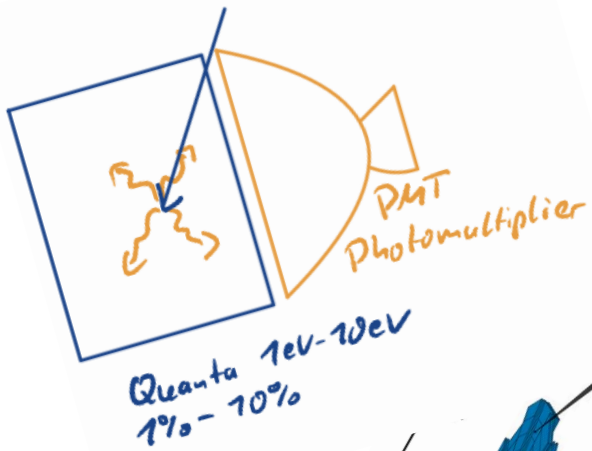


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1	14/10/25	Introduction	Little History of particle physics	atoms, elementary particles
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3	28/10/25		Fundamental symmetries	Conservations, Feynman, Propagator,
4	04/11/25		Fundamental symmetries	Discrete symmetries; Cross section
5	11/11/25		Interaction with Matter	Bethe Bloch, gamma interaction
6	18/11/25		Particle detection	Gamma interactions, detector basics
7	25/11/25	Astrophysics	Detectors, accelerators	Detector types, accelerators, cyclotron Supernovae, cosmic rays, Supernovae, cosmic rays
8	02/12/25	Nuclear physics	Basics	Binding energies, Weizäcker
9	09/12/25		Nuclear reactions	Alpha, beta, gamma
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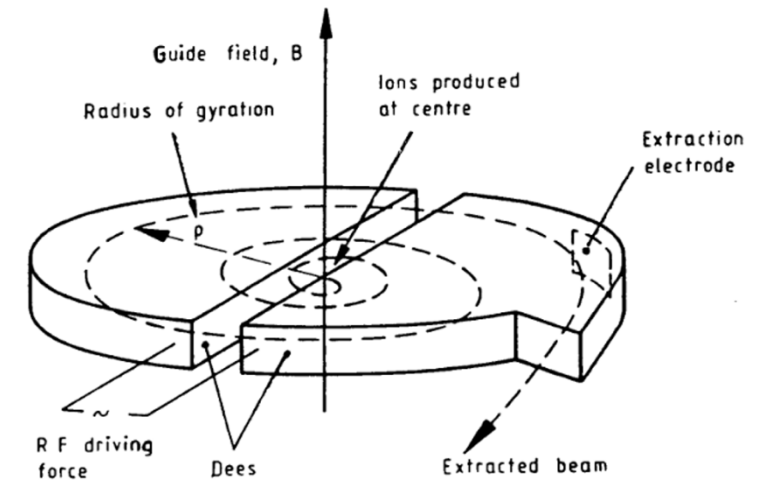
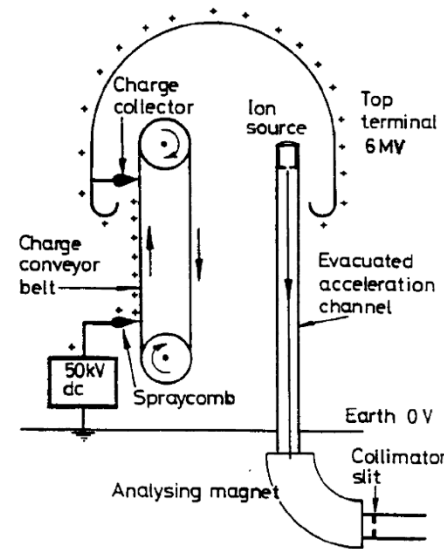


Zusammenfassung der letzten Vorlesung



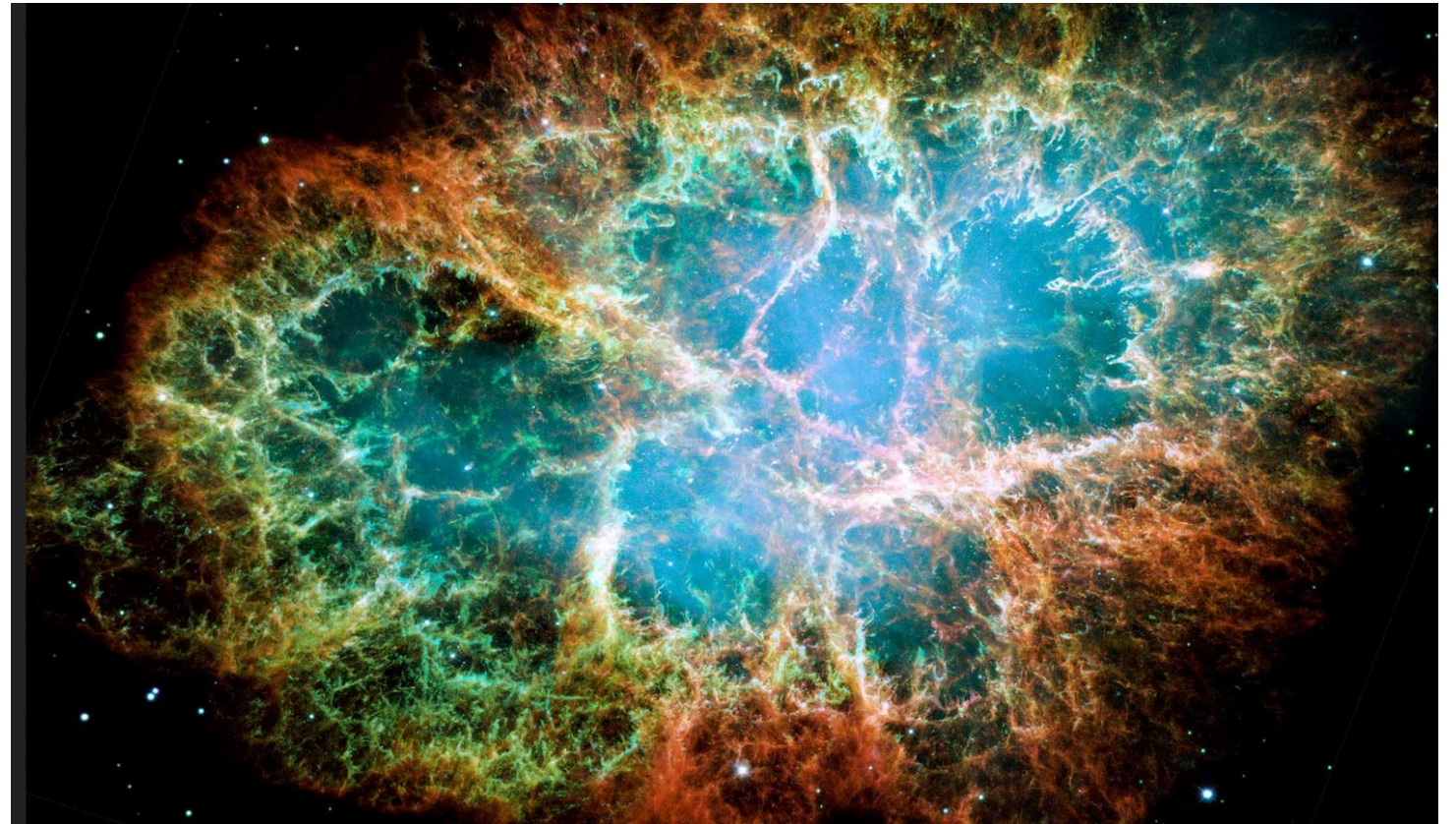
Detektoren : Prinzip und Realität

$$\vec{F}_L = q(\vec{E} + \vec{v} \times \vec{B})$$



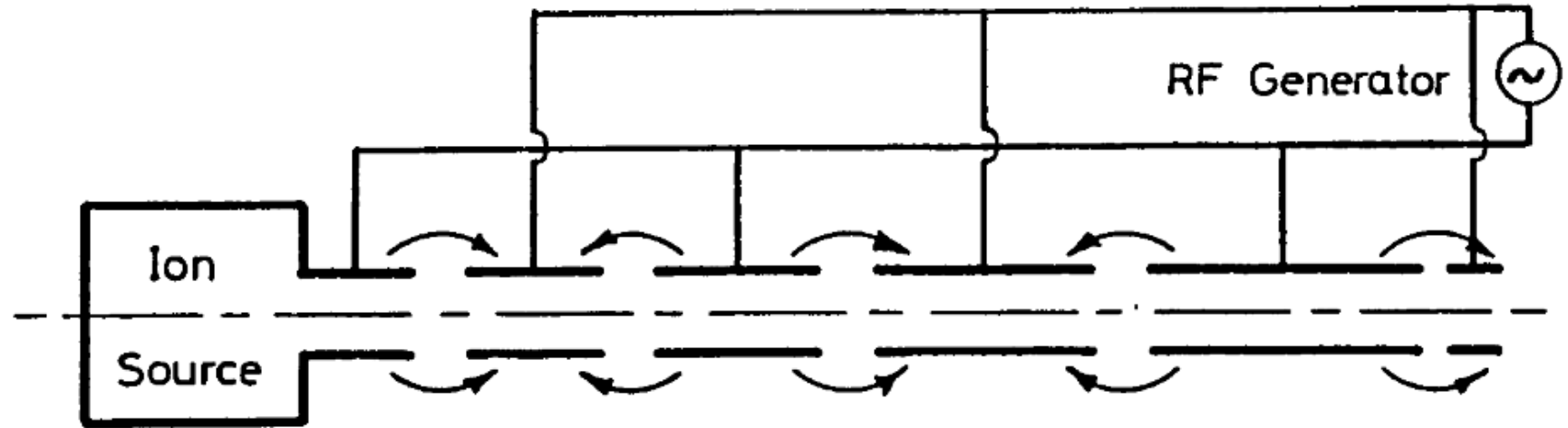
Beschleuniger

Terrestrische und kosmische Beschleuniger



Isings und Wideröes Linearbeschleuniger

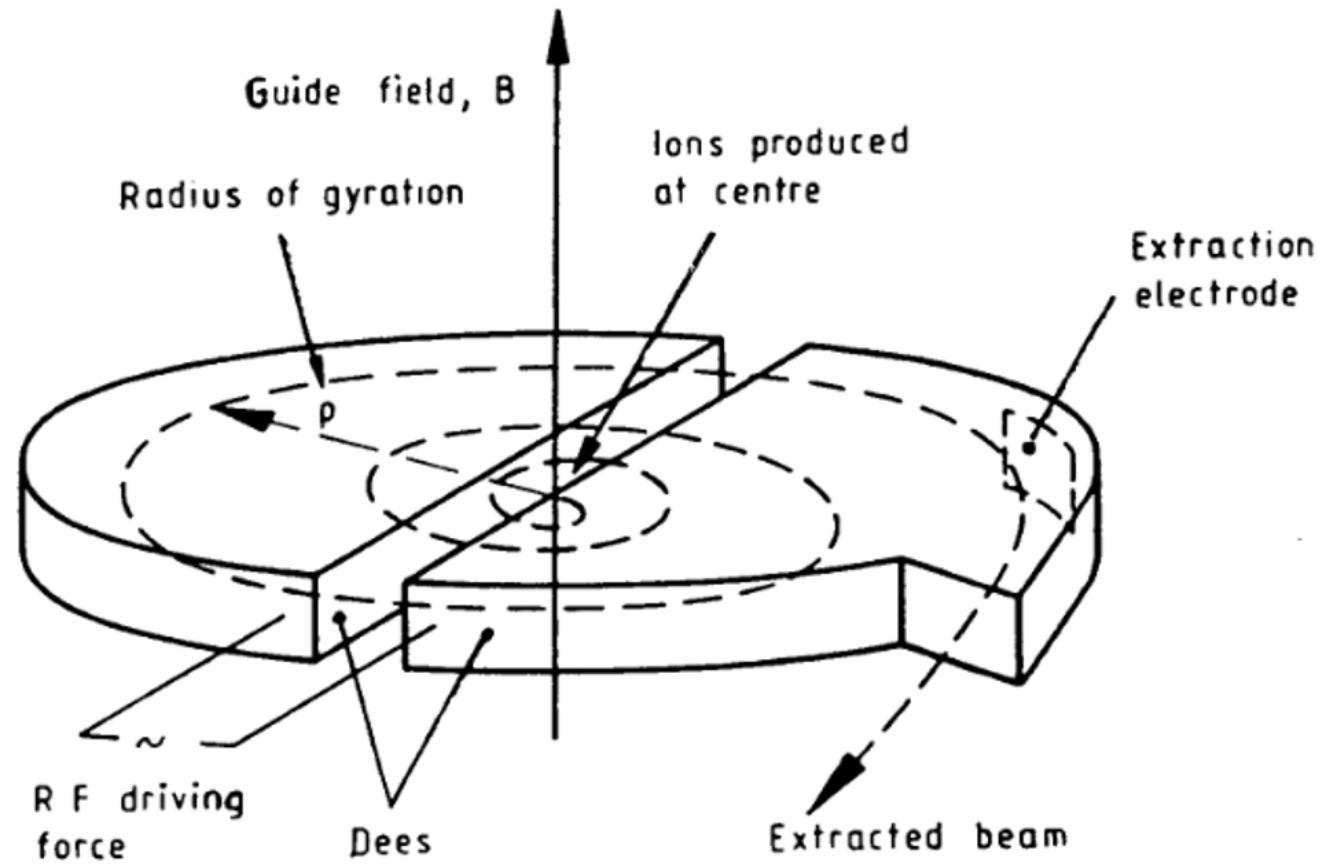
- ▷ lineare Kette von leitenden Driftröhren,
- ▷ an Hochfrequenz angeschlossen,
- ▷ Teilchen entlang Längsachse,



- ▷ Hochfrequenz synchron zum Teilchenflug:
Teilchen zwischen Driftröhren, wenn beschleunigendes $\vec{E}$ -Feld, sonst im feldfreien Raum einer Driftröhre
- ▷ Driftröhrenlänge wächst mit Teilchengeschwindigkeit

Wideröe hat 1928 den ersten funktionierende Beschleuniger nach Isings Vorschlag gebaut.

Zyklotron



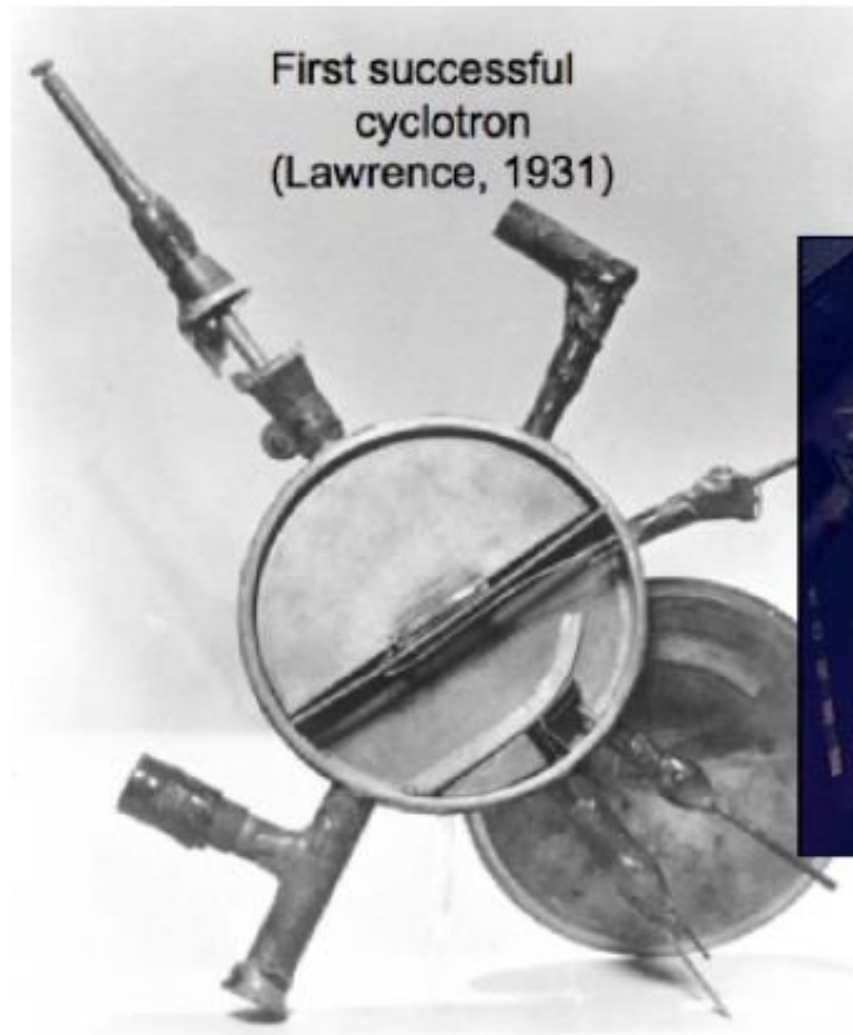


Figure 4.6: *A first prototype of a cyclotron (by Lawrence) and the 590 MeV isochronous cyclotron at PSI.*

Synchrotron

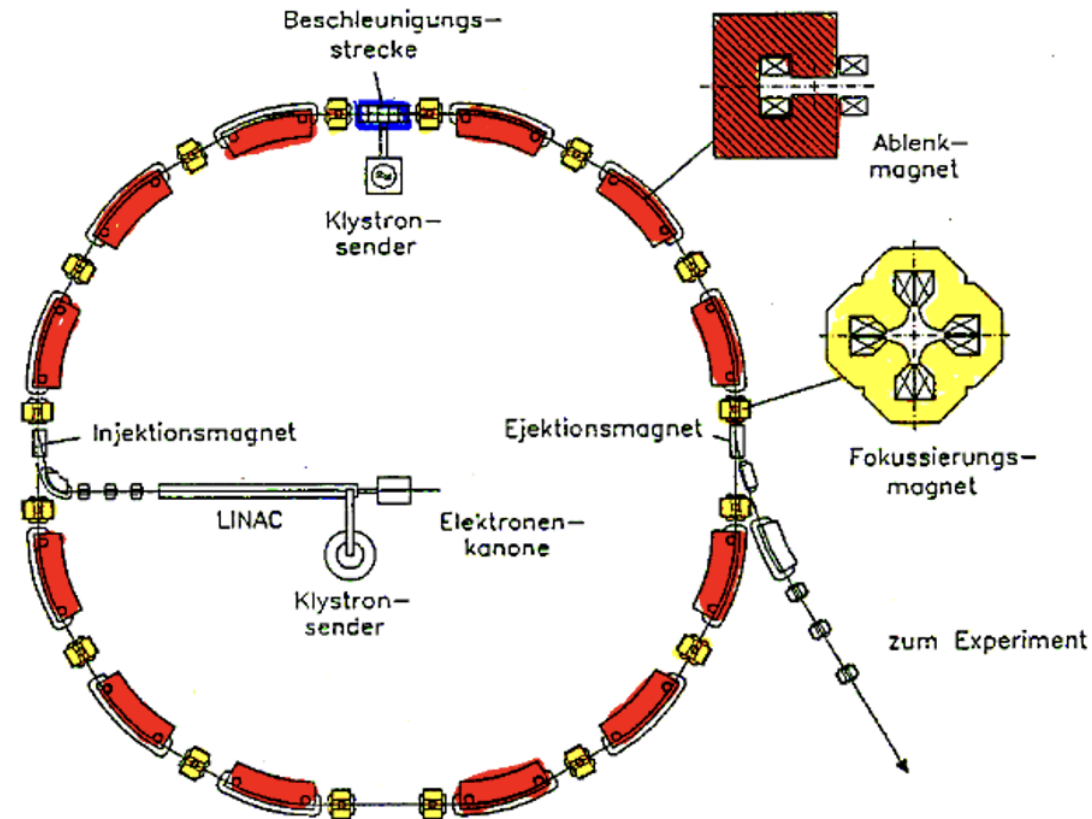
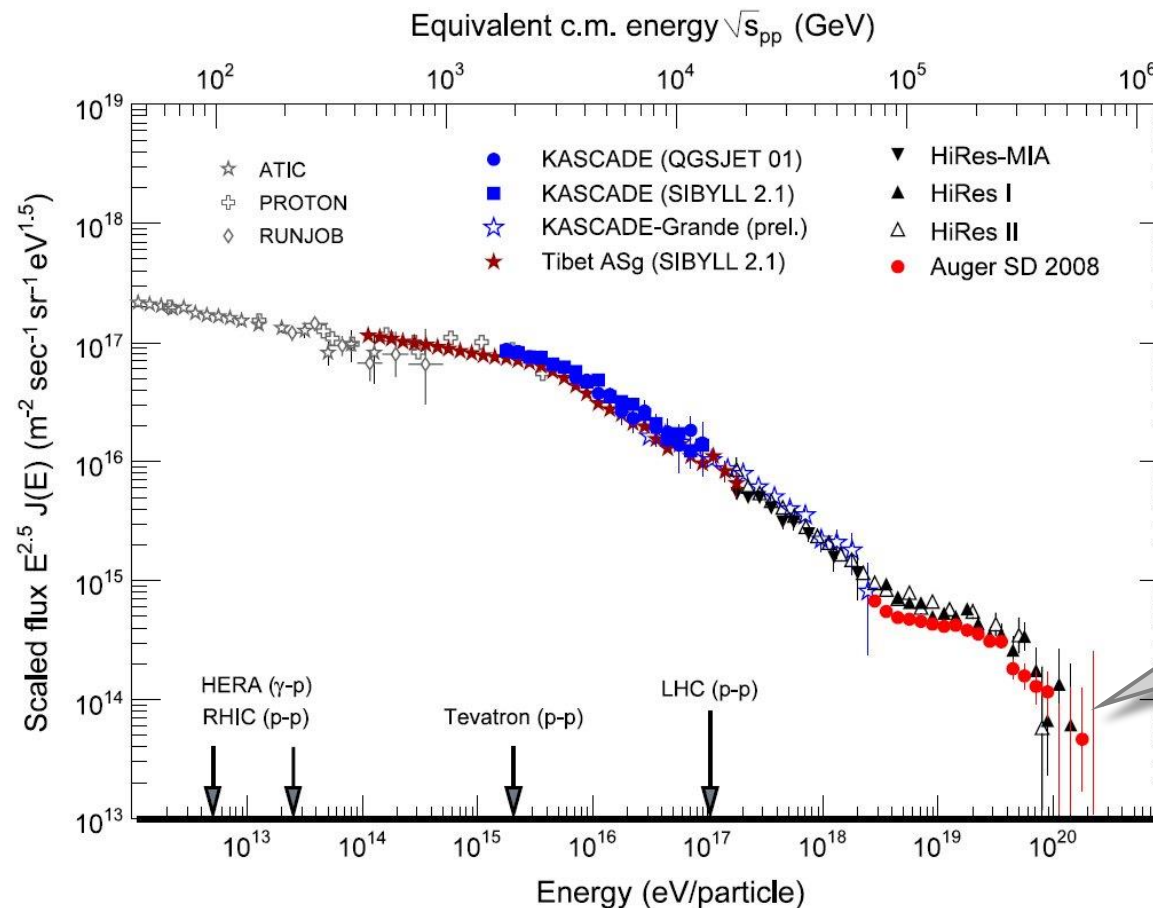


Fig. 1.16 Prinzipieller Aufbau eines modernen Synchrotrons. Die Bahn wird durch Ablenk-magnete mit homogenem Feld festgelegt, während die Fokussierung des Strahls durch geson-derde Magnete besorgt wird. Die Beschleunigung geschieht durch eine oder mehrere kurze HF-Strukturen. Die Teilchen werden von einem Vorbeschleuniger (Linac oder Microtron) ge-liefert.

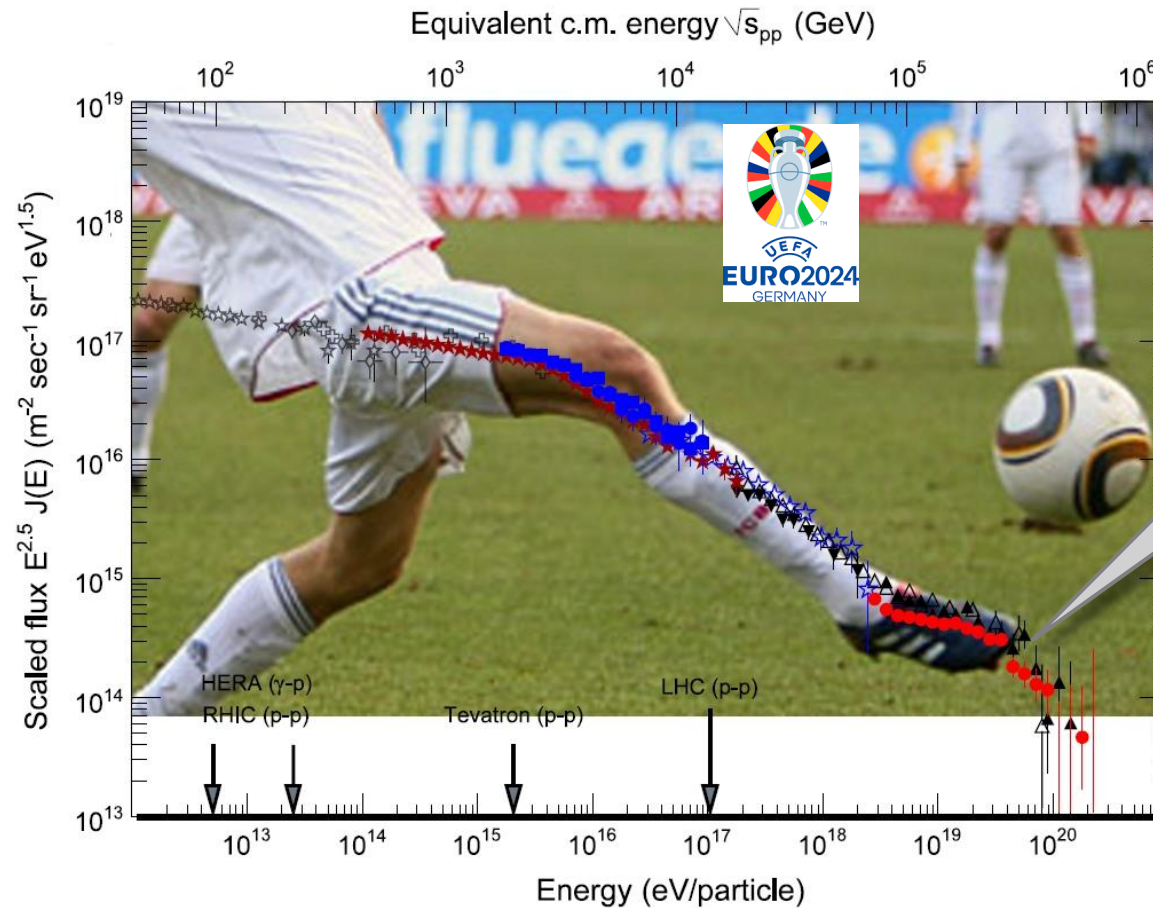
The energy spectrum of Cosmic Rays



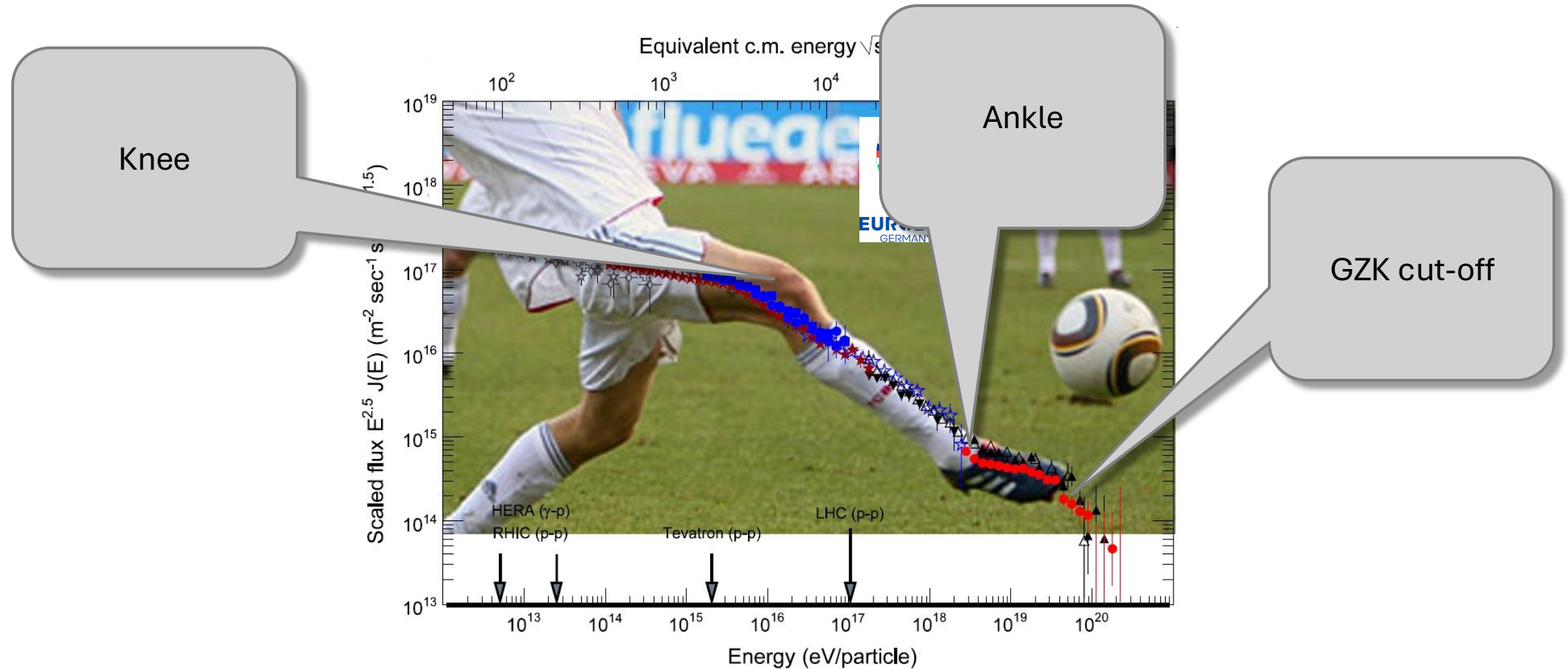
Center of mass
energy:
1000 TeV

Energies many
orders of
magnitude
higher than at
LHC

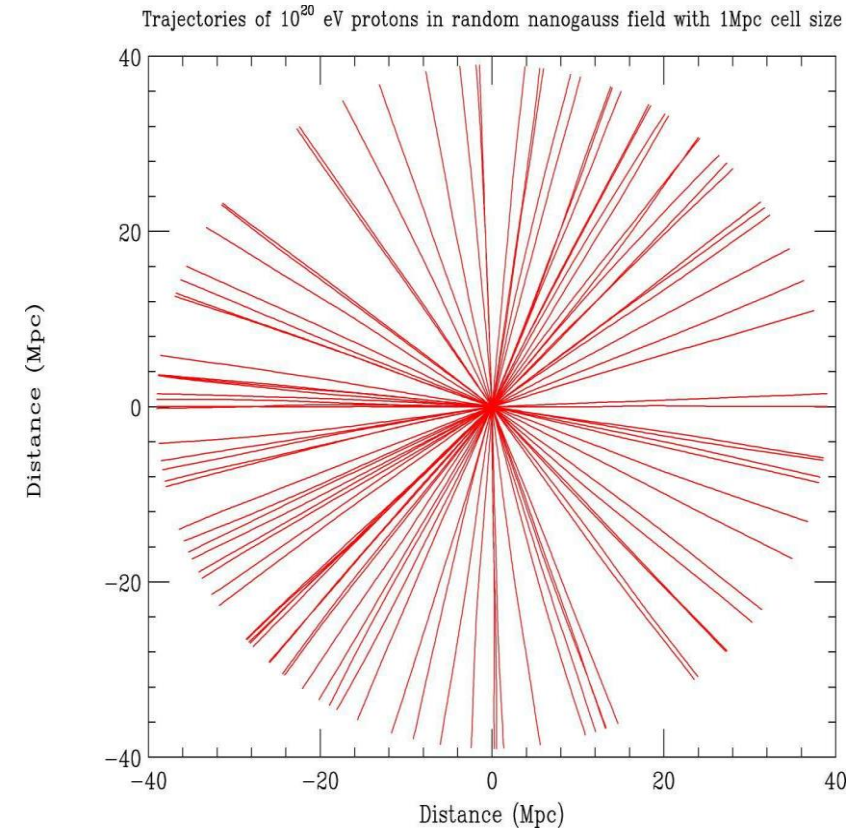
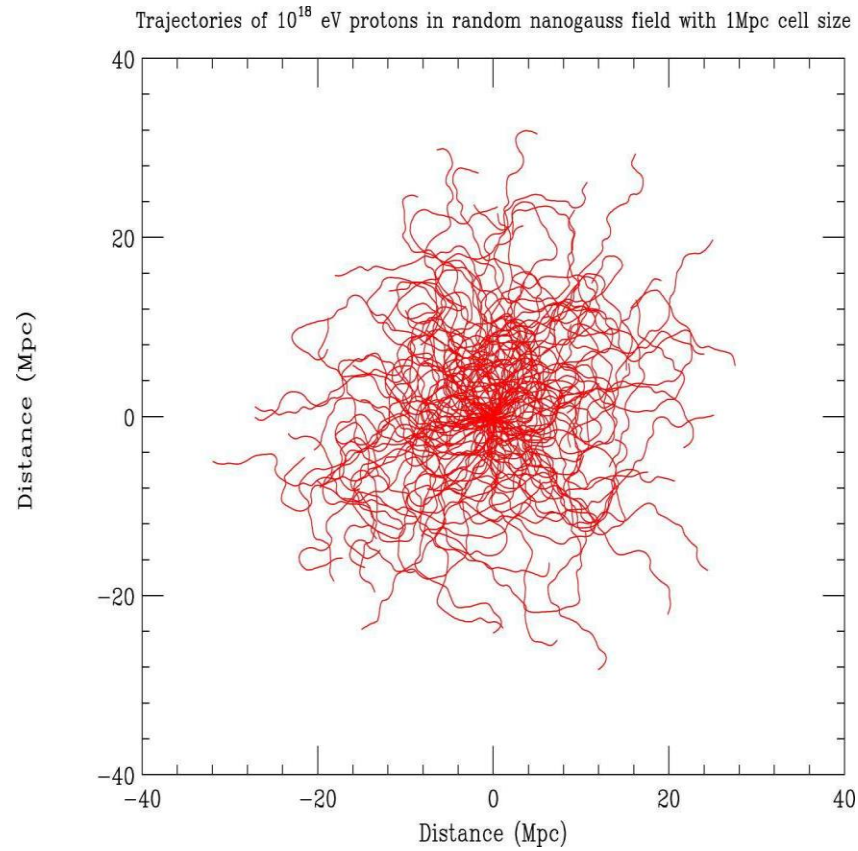
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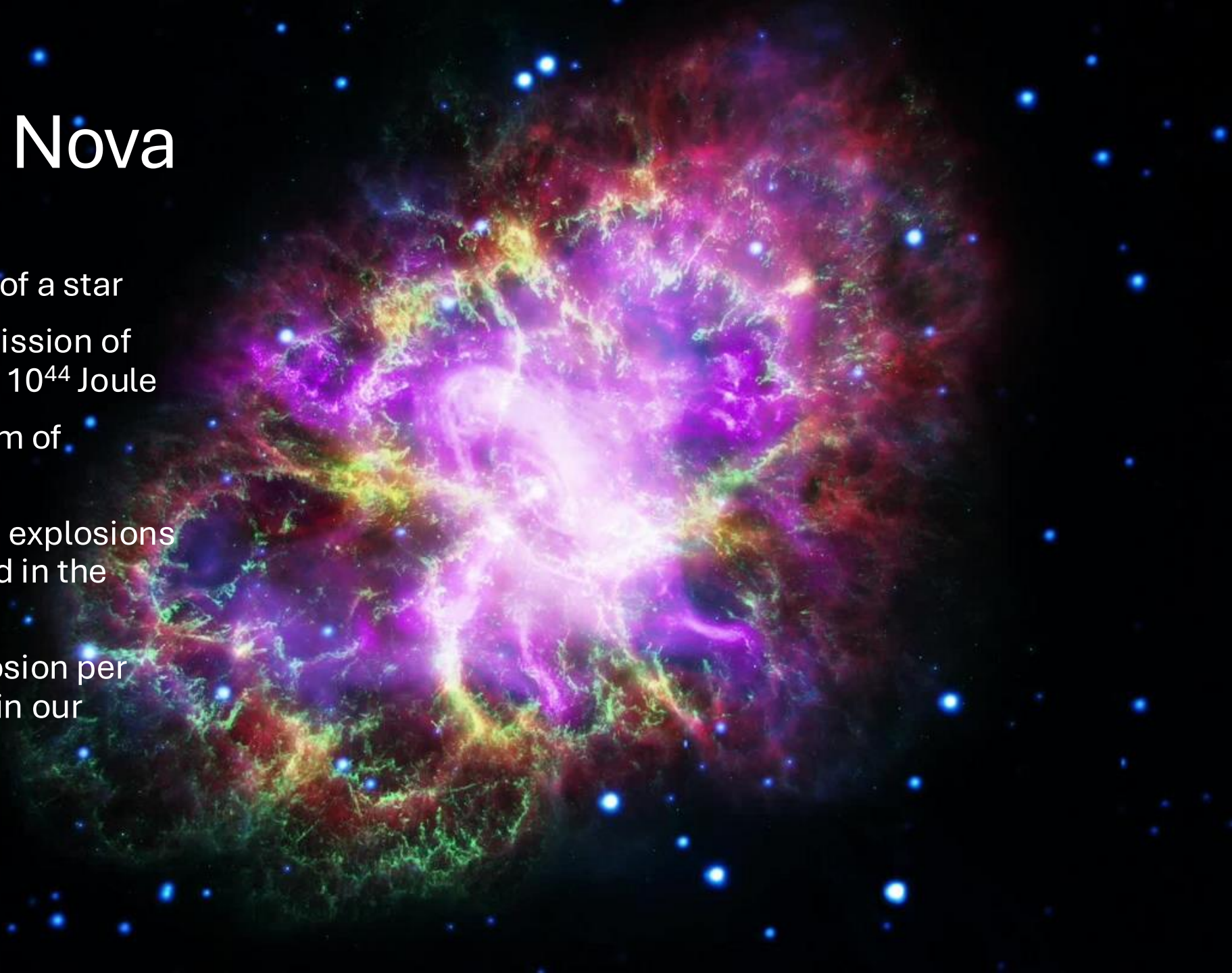


Energien höher als 10^{18} eV werden nicht mehr vom galaktischen Magnetischen Feld in der Galaxie gehalten



Super Nova

- Explosion of a star
- Energy emission of 10^{51} erg or 10^{44} Joule
- 99% in form of neutrinos
- 20 – 30 SN explosions per second in the universe
- 3 SN explosion per 100 years in our galaxy



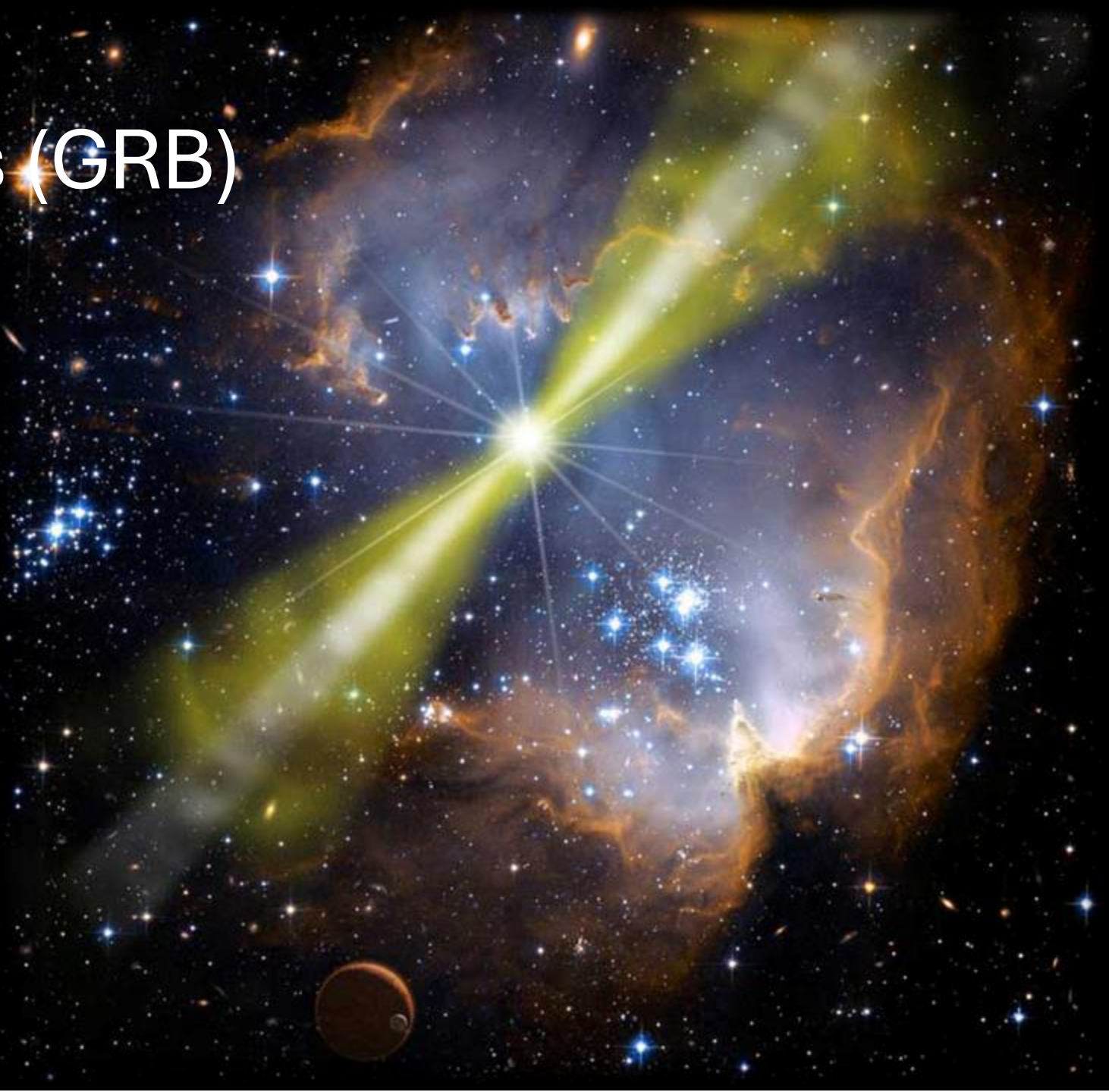
Active Galactic Nuclei (AGN)

- Active galactic nuclei (AGNs) are compact regions in the centers of galaxies that appear very bright in electromagnetic radiation
- Regions contain supermassive black holes that accrete surrounding matter, converting gravitational energy into radiation
- In-falling matter forms an accretion disk. In many AGNs, relativistic jets perpendicular to the disk are observed
- Example: Quasar, Blazar



Gamma Ray Bursts (GRB)

- GRBs are extremely energetic explosions observed in distant galaxies
- For a period of 10^{-3} to 10^3 seconds they outshine every other source of gamma rays in the Universe
- They were first detected by a set of military satellites, operated by the US Air Force
- Long bursts (duration > 2 s): associated with the core-collapse of a supermassive star
- Short bursts (< 2 s): result of two merging neutron stars, or the merging of a neutron star and a black hole



Greisen–Zatsepin–Kuzmin (GZK) Cut-off

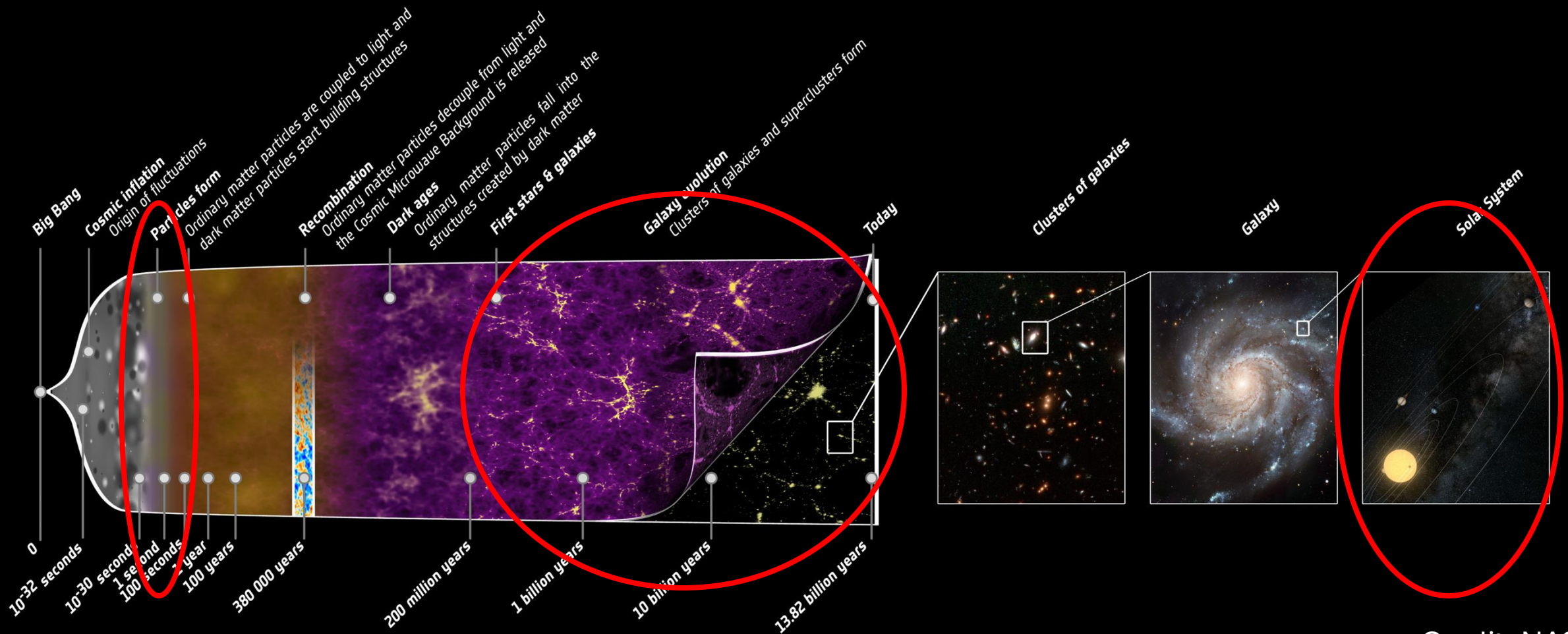


- Slowing interactions of CR particle with the CMB
- $\gamma_{CMB} + p \rightarrow \Delta^+ \rightarrow p + \pi^0$
- $\gamma_{CMB} + p \rightarrow \Delta^+ \rightarrow n + \pi^+$

Kernphysik

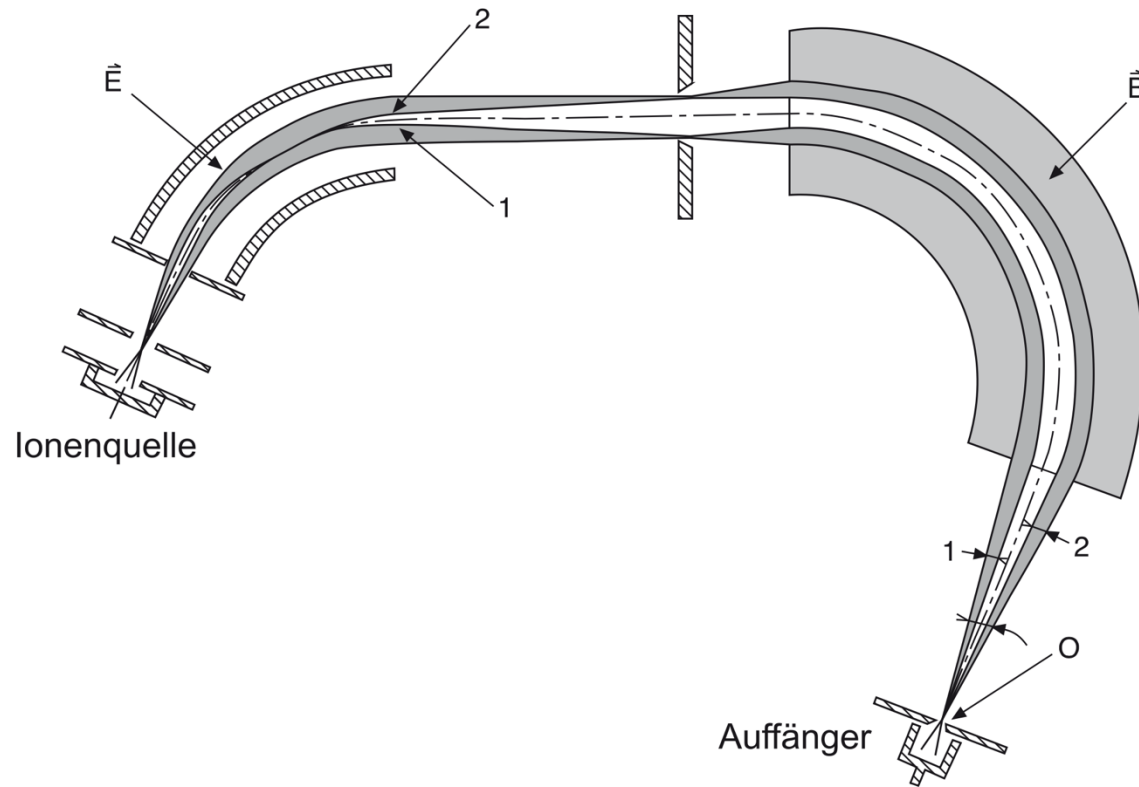
Entstehung von...

- leichten Kernen im frühen Universum (10^{-10} bis 10^3 sek. nach dem Urknall)
- Kerne bis Fe in stellaren Brennen (Fusion)
- schwere Kerne in Supernova und anderen Orten mit extrem hohen Neutronenflüssen



Credit: NASA

Massenbestimmung durch Massenspektroskopie



$$r_E = \frac{M}{Q} \cdot \frac{v^2}{E}$$

$$r_M = \frac{M}{Q} \cdot \frac{v}{B}$$

Radien r_E (Coulombkraft im E-Feld) und r_B durch Lorentzkraft im B-Feld

Alternativ/zusätzlich kann man Kernmassen über Kernreaktionen ermitteln:

